Linear algebra review materials

This course is about linear models. Furthermore it is, at least in part, a math course. So a strong foundation in linear algebra will be necessary. We will do our best to provide support to either learn or review linear algebra, but there is no substitute for working through the material yourself.

Below are some materials that I can recommend for self-study and / or review based on the MIT open courseware linear algebra series. The corresponding textbook, Strang's Introduction to Linear Algebra, is also good, but is unfortunately not available for free online, and there are only a few hard copies available in the library.

Not all the material in the MIT linear algebra is necessary for 151A. Based mostly on the titles, here are the sections of the reading that I recommend for 151A:

- Absolutely necessary: Sections 3, 6, 10, 14
- Important: Sections 7, 15, 16, 10 (video 9), 11 (video 9)
- Very useful: Sections 17, 21, 27 (video 25), 28 (video 25), 33 (video 30)

Note that the numbers in the recommended reading don't seem to line up with the videos, particularly later in the course. Where it seems the two diverge, I've tried to guess which video goes with the corresponding reading.

As is always the case with math, reading and listening is no substitute for doing.
Actually working through practice problems is the key to making linear algebra second nature! So if you have a choice between, say, practicing doing some matrix multiplication yourself and watching another video on the singular value decomposition, I'd recommend the former until you're very comfortable with the basics.